

Chapter 165 — Management of Recurrent and Newly Metastatic Prostate Cancer

- **Biochemical recurrence (BCR)** is progression **unique to prostate cancer**: after curative-intent therapy a **prostate-specific antigen (PSA) rise precedes clinical recurrence** (recurrence **detectable on imaging**).
- Management depends on **prior therapy, patient goals, and risk of future clinical recurrence** — **further curative therapy** for some, **systemic control** for others; progression to metastasis is managed with **combination systemic therapy**.

CURATIVE INTENT APPROACHES TO BIOCHEMICAL RECURRENCE

After Radical Prostatectomy

Definition of Biochemical Recurrence After Radical Prostatectomy

- **About a quarter to two-fifths** develop PSA recurrence **within 10 years** of radical prostatectomy (RP); **high-risk disease approaches 60% at 5 years**. The **2012 US Preventive Services Task Force grade D** recommendation has increased **high-risk disease at diagnosis**, and hence potential relapse.
- **A detectable PSA is not destiny**: about **half** of intermediate/high-risk men with detectable PSA ever reach **clinical failure** (metastasis, second-line treatment, or cancer death). **Why: residual benign tissue** (the usual explanation after radiotherapy), **non-prostatic PSA**, or **indolent low-grade residual cancer**.
- **The governing asymmetry: not every detectable PSA precedes clinical recurrence, but clinical recurrence is almost always preceded by a rising PSA** — hence the search for a threshold that predicts progression early enough to maximise benefit from further therapy.
- **PSA ≥ 0.4 ng/mL followed by a further rise** was most associated with **metastasis**, secondary therapy, further PSA progression and rapid doubling time (Memorial Sloan Kettering [MSK] series).
- Any fixed threshold is complicated by **ultrasensitive assays, clinical risk features**, and the **effect of PSA on salvage outcome**.

Ultrasensitive Prostate-Specific Antigen

- Assays reach **0.01 ng/mL**; earlier detection brings **anxiety and uncertainty**, without high-level evidence that treating on these values helps.
- **Prognostic, not diagnostic**: an **undetectable nadir predicts freedom from BCR**; **each month undetectable lowers risk**; the value **at 3 years** predicts eventual BCR.
- **Pathology modifies meaning**: discrimination is sharp in **pT3 disease** but **negligible in low-risk disease** → relevant mainly to **intermediate/high risk**.
- About **half of early (1–3 month) detectable values fall** on retesting → **closer follow-up, not treatment**.

Clinical Factors Associated With Recurrence

- Context changes meaning: **0.1 ng/mL differs 8 years after grade group 1 versus 6 months after grade group 5**.
- A **single PSA ≥ 0.1 ng/mL** predicts further therapy in **most men with high-risk pathology (pT3b, grade group 4–5)** but **few with favourable pathology (pT2, grade group 1–2, negative margins)**.
- **Positive margins, grade group, T stage, PSA doubling time and pre-salvage PSA** predict metastatic recurrence (meta-analysis) — effect sizes too heterogeneous to pool.

Effect of Prostate-Specific Antigen on Salvage Radiation

- **Lower PSA at salvage radiotherapy (RT) predicts freedom from recurrence**, even below 0.5 ng/mL.
- **Adverse pathology amplifies the PSA effect**: with ≥ 2 adverse features (positive margin, pT3b, grade group 4–5) **each 0.1 ng/mL costs ~10%** in recurrence risk; with 0–1, **~1.5%**.

*Guideline-Based Definitions of Recurrence***COMPARISON — GUIDELINE DEFINITIONS OF BCR AFTER RADICAL PROSTATECTOMY**

Body	Definition	Approach
American Urological Association (AUA)	PSA ≥ 0.2 ng/mL	Single threshold
National Comprehensive Cancer Network (NCCN)	Failure to reach undetectable, two consecutive rises, or PSA ≥ 0.1 ng/mL	Any of three triggers
European Association of Urology (EAU)	No absolute threshold. Low-risk BCR: PSA doubling time >1 year AND grade group <4 . High-risk BCR: doubling time ≤ 1 year OR grade group 4–5	Risk categories

- **What it tests**: AUA and NCCN define recurrence **by a number**; EAU **by kinetics and grade**. **Trap**: AUA (0.2) is **higher** than NCCN (0.1); EAU low risk needs **both** favourable features, high risk **either** adverse one.
- Whatever the definition, the aim is to find **men who will benefit**, weighing biochemistry, pathology, goals and **overtreatment**.

Prediction of Postprostatectomy Biochemical Recurrence

- Needed for **counselling, early salvage and trial selection**. Established factors: **pre-treatment PSA, clinical stage, pathological grade group, pathological stage, margin status**.
- **Positive margin — the only modifiable factor** and one of the strongest.
 - Raises BCR at every pT stage; **pT3a margin-negative** carried **lower risk** than **pT2 margin-positive**; confirmed by meta-analysis.
 - **Beyond binary status**: **length >1 mm, multifocality, location, and Gleason ≥ 4 at the margin**; after adjustment for PSA, stage and grade, **each 1 mm of margin length and Gleason 4–5 at the margin** independently predict BCR and metastasis → **meticulous dissection**.
- **Magnetic resonance imaging (MRI)**: tumour volume, Gleason score, stage and margin status predict BCR; **tumour contact length >14.1 mm with PSA >10 and grade group 3–5** identifies the highest risk.
- **Prostate Health Index (PHI)** (total PSA, free PSA, $[-2]\text{proPSA}$) predicts BCR.
- **Genomic classifiers — Decipher, Genomic Prostate Score (formerly Oncotype DX GPS), Prolaris** — validated for **metastatic recurrence**; **prognostic**, but their role in **selecting early salvage** is unproven.
- **Nomograms**: a **preoperative** tool (stage, PSA, Gleason, systematic biopsy) predicts progression (PSA ≥ 0.4 confirmed, secondary therapy, clinical recurrence, or positive frozen-section nodes) with **concordance 0.76–0.79**; a **postoperative** tool (adding extracapsular extension, seminal vesicle, nodal and margin status, adjuvant RT) predicts **10-year BCR-free survival** with **concordance 0.79–0.81**; University of California, San Francisco (UCSF) **CAPRA** (pre-op) and **CAPRA-S** (post-op) are validated for **5-year BCR**.

TABLE 165.1 ONLINE RISK STRATIFICATION NOMOGRAMS FOR PROSTATE CANCER

Setting	Nomogram	Variables	Outcome
Pre-treatment	MSK Pretreatment	PSA, age, biopsy Gleason, stage, number of positive biopsies	5-, 10-year BCR; 15-year cancer-specific survival (CSS)

Setting	Nomogram	Variables	Outcome
	Stephenson	Preoperative PSA	10-year BCR
	UCSF Cancer of the Prostate Risk Assessment (CAPRA)	Age, PSA, biopsy Gleason, stage, % cores positive	2-, 3-, 5-year BCR
Post-treatment	MSK Post-treatment	PSA, age, pathological Gleason, era, months cancer-free, margins, extracapsular extension, seminal vesicles, nodes, hormones, prior RT	2-, 5-, 7-, 10-year BCR; 15-year CSS
	UCSF CAPRA-S	PSA, margins, seminal vesicle invasion, Gleason, extraprostatic extension, nodes	3-, 5-year progression-free survival

- **What it is for:** **biopsy-based** tools before treatment, **pathology-based** tools after; only **MSK** tools reach **15-year CSS**. Trap: CAPRA pre-op, CAPRA-S ("surgery") post-op.

Natural History of Biochemical Recurrence Postprostatectomy

- BCR precedes metastasis and death — slowly and variably.
- **Pound series (1999, seminal):** median **8 years BCR** → metastasis, **5 years metastasis** → death; risk depended on **time to BCR**, **Gleason score**, **PSA doubling time**. An update: **median metastasis-free survival (MFS) 10 years**, reflecting better selection and management.
- **Most men with BCR do not die of prostate cancer;** even at intermediate/high risk about **half never fail clinically**. When they do, **5-year disease-specific survival falls with site: local > nodal > distant**. Even **node-positive at RP**, only about half of those with BCR fail clinically.
- **Nomogram BCR-risk quartile predicts death:** 15-year disease-specific mortality is several-fold higher in the least favourable quartile.
- **PSA follow-up is lifelong** — up to about **a quarter recur after 5 years**, still at risk of progression.

KEY POINTS

- BCR is the **intermediate state** of recurrence, **defined by PSA**, and guides further treatment in selected patients.
- **AUA: PSA ≥ 0.2 ng/mL; NCCN: failure to reach undetectable PSA, two consecutive rises, or PSA ≥ 0.1 ng/mL.**
- Risk is assessed by **clinicopathological features** (PSA, stage, grade group, margins), **genomics** and **nomograms**.
- BCR **does not universally lead to metastasis or death**; act in the context of **patient goals and longevity**.

Evaluation of Postprostatectomy Biochemical Recurrence

- The curative option is **salvage RT to the pelvis**, so evaluation aims to **exclude metastatic (non-local) recurrence**, where local therapy is **risk without benefit**.
- **Conventional imaging fails at low PSA:** bone scan and **computed tomography (CT)** rarely positive **below PSA 10 ng/mL**; **^{18}F -sodium fluoride positron emission tomography/CT (^{18}F -NaF PET/CT)** beats planar bone scan but **equals bone scan plus single-photon emission CT (SPECT)**; **transrectal ultrasonography (TRUS)** detects local masses (especially at the **vesicourethral anastomosis**) but **cannot exclude metastasis** → use only if local confirmation would change salvage.

- **Historical only: indium-111 capromab pendetide (ProstaScint) and ¹¹C-acetate. Current: choline, ¹⁸F-fluciclovine, prostate-specific membrane antigen (PSMA). Choline improved on fluorodeoxyglucose (FDG) PET, which is limited by prostate cancer's low glucose metabolism.**

COMPARISON — PET TRACERS IN BIOCHEMICAL RECURRENCE

Tracer	Approval	Distinguishing features	Low-PSA performance
¹¹ C-choline	FDA (US Food and Drug Administration)-approved	Low urinary excretion → good pelvic images; short half-life needs on-site cyclotron	~Half positive below 1.0 ng/mL; PSA ≥1 and doubling time <6 months predict metastasis on imaging
¹⁸ F-choline	—	Higher urinary excretion, longer half-life → no cyclotron needed	Similar to ¹¹ C
¹⁸ F-fluciclovine	2016	LOCATE: recurrence found in a quarter at median PSA 0.57; EMPIRE-1 (phase II/III): fluciclovine-guided salvage improved 3-year control vs conventional imaging (75.5% vs 63.0%)	Poor below 0.5; none below 0.2
⁶⁸ Ga-PSMA-11	2020	Highest detection	~Half positive below 0.5; ~100% above 2
¹⁸ F-piflufolastat (PSMA)	2021	¹⁸ F label	~Half positive below 0.5, some extrapelvic

- **Pattern: detection rises with PSA for every tracer; pooled detection below 0.5 ng/mL: PSMA > fluciclovine. → PET/CT with an approved agent before curative salvage, to avoid irradiating the pelvis in metastatic disease.**
- **MRI for local recurrence: endorectal MRI is highly accurate — recurrence isointense to muscle on T1, slightly hyperintense on T2; dynamic contrast enhancement (DCE) raises accuracy at lower PSA; DCE + magnetic resonance spectroscopic imaging (MRSI) stays sensitive at PSA 0.4–1.4. Multiparametric MRI (mpMRI) beats PET/CT for local recurrence, especially at low PSA, and a visible local recurrence predicts benefit from salvage RT.**

PAIR — WHAT EACH MODALITY IS FOR AFTER PROSTATECTOMY

PET/CT (choline, fluciclovine, PSMA)	mpMRI
Excludes metastatic disease; beats CT and bone scan for distant disease	Finds local recurrence; beats PET/CT locally, especially at low PSA
Selects men for local salvage	Guides salvage RT

- **Fig. 165.1 — ¹⁸F-piflufolastat PET/CT showing a small internal iliac nodal recurrence invisible on conventional CT.**

KEY POINTS

- Imaging in BCR aims to rule out metastasis before further local therapy.
- CT and bone scan have limited utility at low PSA after prostatectomy.
- PET (choline, fluciclovine, PSMA) is superior to CT for metastatic recurrence and helps select for local salvage.
- MRI is superior for local recurrence and can guide salvage RT.

Salvage Radiation Therapy

- About **two in five** men develop BCR after surgery; observation and hormones are options, but **salvage RT is the only curative-intent therapy**.
- **Outcomes:** roughly a **third to a half BCR-free at 4–5 years**, even with high-risk features. **Predictors of durable response:** Gleason grade, pre-treatment PSA, margin status, PSA doubling time, seminal vesicle invasion. Salvage RT sharply reduces **local recurrence**, **later androgen deprivation therapy (ADT)** and **systemic progression**.
- **CSS benefit (~threefold vs observation)** was confined to men treated **within 2 years of BCR** with **doubling time <6 months** — the highest metastatic risk. Median overall survival (OS) after a **second failure** still exceeds a decade.
- **Predicting failure:** a salvage nomogram (PSA, pathological Gleason, doubling time, margins, nodes, ADT) has **concordance 0.69**; **shorter time to failure** predicts metastasis and death. **Margin location and number do not predict failure**, but **size and grade do** — in grade group 3–5, a **margin >3 mm** roughly halves survival.
- **Harms:** **genitourinary (GU)** — cystitis, stricture; **gastrointestinal (GI)** — proctitis, urgency, tenderness; **erectile dysfunction**.
- **When to start:** older data supported **PSA ≤0.5**; **lower is better** — each 0.1 ng/mL costs ~2.6% recurrence-free survival.
- **Early salvage vs adjuvant (landmark):** RADICALS, RAVES, GETUG-AFU 17 — **early salvage at PSA 0.1–0.2 vs adjuvant RT** in high-risk pathology; pre-planned meta-analysis (**ARTISTIC**): **equivalent biochemical control**, ≥60% spared RT → **PSA 0.1–0.2 is a reasonable trigger** in high-risk men.
- **Imaging and timing:** **no visible lesion on mpMRI predicts salvage failure** (possible occult metastases); ⁶⁸Ga-PSMA-11 PET/CT finds disease **outside the planned field** in about one in five at low PSA (commonest **bone, perirectal**). **The inverse relationship:** imaging detection **rises**, and salvage efficacy **falls**, as PSA rises.

Salvage Radiation Therapy Dose

- Borrowed logic from primary RT (**higher dose, better control**); **intensity-modulated radiotherapy (IMRT)** replaced conformal RT. Series and a meta-analysis favour doses **>66 Gy** (~2% recurrence-free survival per Gy), escalation to 75–76 Gy with acceptable toxicity — but the randomised **SAKK 09/10** found **no gain from 70 Gy** (table below). **ASTRO (American Society for Radiation Oncology)/AUA:** 64–65 Gy minimum.

Whole Pelvis Versus Prostatic Bed Radiation Therapy

- **Prostate bed only**, or **bed + pelvic nodes (whole pelvis)**. Retrospective data were mixed: whole pelvis helped **high-risk features** (Gleason ≥8, PSA >20, seminal vesicle, capsular or nodal involvement), **Gleason ≥7**, and men with **pre-RT PSA ≥0.4**. **Best evidence:** **RTOG 0534 (SPPORT)** (table below) — **Figs. 165.2 and 165.3** show the two fields.

Concurrent Androgen Deprivation Therapy With Salvage Radiation

- **Rationale:** sterilise micrometastases, shrink tumour, radiosensitise.

COMPARISON — RANDOMISED TRIALS IN SALVAGE RADIOTHERAPY

Trial	Axis	Population	Arms	Result	Verdict
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Timing	High-risk pathology	Adjuvant vs early salvage (PSA 0.1–0.2)	Equivalent control; ≥60% spared RT	Early salvage

Trial	Axis	Population	Arms	Result	Verdict
SAKK 09/10	Dose	Median PSA 0.3	64 vs 70 Gy	Freedom from recurrence 62% vs 61% ; late GU/GI toxicity 30%/11%	64 Gy sufficient
RTOG 0534 (SPPORT)	Field + ADT	High-risk pathology, PSA ≥ 0.1 to < 2.0	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4% ; no subgroup failed to benefit from ADT	Add ADT; add pelvic nodes
RTOG 9601	ADT	PSA 0.2–4.0 ≥ 8 weeks post-RP; pT3N0 or pT2 margin-positive	RT 64.8 Gy ± 2 years bicalutamide	12-year OS 76.3% vs 71.3% ; cancer death 5.8% vs 13.4% ; metastasis 14.5% vs 23%	OS benefit
GETUG-AFU 16	ADT	Undetectable then rising PSA 0.2–2.0	RT ± 6 months goserelin	10-year BCR-free 64% vs 49% (HR 0.54); MFS 75% vs 69% (HR 0.73); OS unchanged	BCR and MFS benefit only

- **Pattern:** long antiandrogen gave OS benefit (RTOG 9601); short luteinising hormone–releasing hormone (LHRH) agonist did not (GETUG-AFU 16); SPPORT shows ADT and pelvic nodes are additive. Trap: SPPORT's endpoint was **freedom from progression**, not OS.
- **AUA:** offer hormonal therapy to men with PSA recurrence who are **salvage RT candidates**.

Follow-up After Salvage Radiation Therapy

- **No guideline.** Roughly **one in eight to one in three** suffer a **second BCR** → continue **PSA surveillance** with restaging on recurrence.

KEY POINTS

- **Salvage RT is the only curative-intent therapy** for post-prostatectomy BCR.
- **Earlier RT gives better outcomes; early salvage (PSA 0.1–0.2) may equal adjuvant therapy.**
- **Dose ≥ 64 –65 Gy to the fossa and 40–45 Gy to pelvic nodes.**
- **Adding ADT improves survival.**

Concept — After prostatectomy two curves run in opposite directions: salvage radiotherapy works best at the lowest PSA, but imaging and the certainty that recurrence is real improve only as PSA rises. Risk features — grade, margins, kinetics — are what let the clinician act early without overtreating.

AFTER DEFINITIVE RADIATION THERAPY

Definition of Biochemical Recurrence After Radiotherapy

- After radiotherapy a remnant gland keeps making PSA, so surgical thresholds fail.
- **Phoenix definition (ASTRO/Radiation Therapy Oncology Group [RTOG], 2005):** PSA ≥ 2 ng/mL above nadir ("nadir + 2") — designed to apply **with concurrent ADT**; validated as the **strongest predictor of metastasis, disease-specific and overall mortality** after adjustment; used in modern trials.
- **Brachytherapy (BT) exception:** BT gives ≥ 2 -fold the dose of external beam radiotherapy (EBRT), so Phoenix may **flatter BT**. After **low-dose-rate (LDR) BT**, a ≥ 0.4 ng/mL threshold detects more failures **~12 months earlier**; **ASCENDE-RT (EBRT vs EBRT + LDR-BT boost)** retained its BT advantage under a **surgical definition (≥ 0.2)**; PSA < 0.2 at 4 years after LDR-BT defines **cure** in almost all.

COMPARISON — BCR DEFINITIONS BY PRIMARY TREATMENT

	Radical prostatectomy	EBRT ($\pm$ ADT)	LDR brachytherapy
Why it differs	No gland — any PSA suspect	Remnant gland makes PSA	$\geq 2\times$ dose $\rightarrow$ very low nadir
Definition	AUA ≥ 0.2 ; NCCN ≥ 0.1 / non-nadir / two rises; EAU risk groups	Phoenix: nadir + ≥ 2	≥ 0.2 may suit better; <0.2 at 4 years = cure

Postradiation Prostate-Specific Antigen Bounce

- **Transient rise of 0.1–0.5 ng/mL** (definitions vary) **then fall**, at a mean ~ 9 months; attributed to **post-radiation prostatitis** or **delayed death of sublethally irradiated cells**.
- **Favourable**: associated with **better BCR-free survival**, and after dose-escalated EBRT with **lower PSA recurrence, metastasis and all-cause mortality** $\rightarrow$ **monitor closely; do not salvage**.

Natural History of Prostate-Specific Antigen Recurrence After Definitive Radiation

- PSA-only recurrence leads to **local failure, distant metastasis (more often than local)** and **cancer death** in a substantial proportion; after Phoenix failure, median ~ 5.4 years to metastasis and ~ 10.5 years to cancer death.
- **Predictors**: short PSA doubling time, clinical stage, Gleason score, short interval to BCR.
- Unlike surgery, a **remnant prostate exists to guide further local therapy**.

Evaluation of Postradiation Biochemical Recurrence

- **First ask whether he would benefit from any therapy**: many chose RT because of **limited life expectancy or unfitness**, and post-failure survival is long — evaluate only if **actionable**.

Imaging After Biochemical Recurrence Following Radiotherapy

- **Step 1 — exclude metastasis** (if present $\rightarrow$ **systemic therapy**; local therapy would not prolong survival). Principles mirror post-RP imaging, except the **residual gland** alters fossa assessment.
- **Fibrosis and shrinkage defeat conventional imaging**: TRUS $\approx$ **digital rectal examination**; CT useful only at PSA > 20 .
- **^{68}Ga -PSMA PET** was **positive in all Phoenix failures** in one series (mostly **distant or local + distant**); **^{11}C -choline PET/CT** positive in most, a third extrapelvic — **PSA rise above nadir, time since RT and NCCN risk group** predicted distant disease (nomogram index **0.79**).
- **Step 2 — confirm local recurrence**: **only biopsy proves it**. PET lacks spatial resolution; MRI maps extent and guides biopsy.
- **MRI after RT**: **T2 is poor** (diffusely **low T2 signal**; low sensitivity and positive predictive value) $\rightarrow$ **DCE** (vascularity, angiogenesis, permeability) outperforms T2; **diffusion-weighted imaging (DWI)** improves biopsy accuracy and matches **template mapping biopsy** per patient; **full mpMRI (T2 + DWI + DCE)** reaches **area under the curve ~ 0.84** ; **3-Tesla endorectal mpMRI** before salvage RP stages **extraprostatic extension, seminal vesicle invasion and nodes** reasonably.

Biopsy After Radiotherapy

- **Goals**: presence/absence of residual disease and its **grade**. **Sequence**: **targeted lesions $\rightarrow$ seminal vesicles $\rightarrow$ systematic whole-gland sampling** (ultrasound- or MRI-guided). **Only when local salvage is contemplated**.
- **Caveat**: radiation **alters architecture** (possible **dedifferentiation**); **post-RT Gleason grading is unreliable and controversial**.

KEY POINTS

- **Phoenix (nadir + 2 ng/mL)** identifies men at risk of metastatic progression after **EBRT ± ADT**.
- After **brachytherapy**, **PSA ≥0.2 ng/mL** may be the more appropriate definition.
- Evaluate with **metastatic assessment (PET/CT)**, then — if local therapy is possible — **MRI** and **biopsy** for histological confirmation.

Concept — *Radiotherapy leaves the organ in place, and every difference from the post-prostatectomy pathway follows: recurrence is measured from a nadir rather than from zero, a benign bounce must be told apart from failure, and local recurrence can — and must — be proven by biopsy before salvage.*

TREATMENT OF POSTRADIATION BIOCHEMICAL RECURRENCE

- **Prerequisites for every salvage option: histologically confirmed local recurrence, imaging-confirmed non-metastatic disease, excellent performance status, ≥5–10 years' life expectancy, and shared decision-making** — morbidity is **high** and control **lower** than for primary treatment.

Salvage Radical Prostatectomy

- **Longest-studied** but **infrequently used**; **high 10-year disease-specific survival** in selected series. **Outcome tracks pathological stage: organ-confined and isolated extracapsular extension good; seminal vesicle invasion and nodal disease poor.**
- **Historically avoided** because of initial preference for RT (reluctance or contraindication to surgery) and **fibrosis-related morbidity** — major complications in up to half, **rectal injury**, frequent **bladder neck contracture**, high **incontinence**, near-universal **impotence**.
- **Contemporary (including robotic):** about **two-thirds recurrence-free**, fewer complications; but men are selected for **radio-resistant** disease, so 5-year biochemical control is **moderate**, predicted by **stage and grade**. **Robotic vs open: better continence, fewer contractures and rectal injuries.**
- **EAU selection:** life expectancy ≥10 years; pre-salvage PSA <10 ng/mL; biopsy grade group ≤3; no nodal disease/metastasis on imaging; initial cT1–T2. The minority meeting all criteria had **better functional and oncological outcomes**.

Salvage Brachytherapy

- **Re-irradiation for local failure after EBRT** — dose to the gland while **sparing previously irradiated neighbours**.
- **High-dose-rate (HDR):** about half BCR-free at 5 years, negligible grade ≥3 GU and no grade 3–4 GI toxicity. **LDR:** about half free at 10 years; **higher grade and PSA** predict failure; grade ≥3 urinary toxicity ~1 in 6, GI rare. **LDR ≈ HDR** for control.
- **Bottom line:** control and GU toxicity **comparable to salvage surgery**.

Salvage Stereotactic Radiation

- **Stereotactic body radiotherapy (SBRT)** after prior EBRT: **modest short-term control** (2-year BCR-free 40–76% in systematic review; about half at 3 years in the largest multicentre series) with **low toxicity**.

Salvage Cryotherapy

- **Selection:** PSA <10 ng/mL (ideally <4) and **PSA doubling time ≥16 months**.

- **Success lacks a single definition** — mostly **Phoenix**; Cryo On-Line Data (**COLD**) Registry and series give roughly two-thirds control at 3–5 years (whole gland); **focal control declines progressively** to 5 years. **Freedom from systemic hormones** is an alternative endpoint — most avoid hormones at 5 years, and **post-cryotherapy nadir <0.2** strongly predicts avoidance.
- **Morbidity then vs now**: incontinence once **most** (mostly moderate–severe, low satisfaction), now **single-digit** thanks to **universal urethral warming** protecting the sphincter; **erectile dysfunction** remains very common; **urethrorectal fistula** — **the most dreaded** — now **1–3.4%**; also **obstruction, urethral sloughing, stricture, rectal pain, scrotal oedema, haematuria**.

Salvage High-Intensity Focused Ultrasonography

- **High-intensity focused ultrasound (HIFU) ≥2 years after EBRT**. Prospective trial endpoints: **PSA nadir ≤0.5 ng/mL + negative 12-month biopsy** — most biopsied were cancer-free; notable grade 3 toxicity (**rectal fistula, osteitis pubis, refractory haematuria**). Largest series: **~half BCR-free at 5 years (Phoenix)**, tied to pre-treatment risk group; risks **re-treatment, incontinence, stricture, fistula, pubic osteitis**.

TABLE 165.2 ONCOLOGIC OUTCOMES IN SELECTED SERIES OF SALVAGE TREATMENTS FOR RADIORECURRENT PROSTATE CANCER

Modality	Biochemical control	Other oncological outcomes	Morbidity / prognostic features
Salvage RP	37–57% (5–10 years; surgical PSA thresholds)	CSS 83–98%; MFS 77% (10 years)	Major complications ~1 in 5 (rectal injury 2%; 5% historically); historical incontinence 80%; ≥1 pad/day 56% (table) vs ≤1 pad/day 65% (text) ; poor prognosis grade group ≥3, pre-salvage PSA ≥4
Salvage BT (LDR/HDR)	52–71% (5–10 years, Phoenix)	CSS 93–97% (5 years); 78% (10 years)	Grade ≥3 GU 17–24%
Salvage SBRT	55–60% (3–5 years, Phoenix)	MFS 89%; ADT-free survival 69% (5 years)	Grade ≥3 GU 8%; grade ≥2 GU ~21% (cystitis/dysuria); impotence 82%
Salvage cryotherapy	55–71% (3–5 years, Phoenix)	MFS 90% (5 years); CSS 91% (10 years); ADT-free 71% (5 years)	Rectourethral fistula ~1% ; osteomyelitis 2%; retention 4%; nadir <0.4 → BCR-free 76% vs 22% ; favourable: pre-salvage PSA <5
Salvage HIFU	48–49% (3–5 years, Phoenix); negative 1-year biopsy 81%	Salvage-treatment-free 37% (5 years)	Grade ≥3 GU up to 20% ; fistula 2–5% ; osteitis pubis 2–3% ; bladder neck contracture 8% (focal)

Data from Marra et al., 2021; Chade et al., 2011; Ward et al., 2005; Smith et al., 2020; López et al., 2019; Fuller et al., 2020; Pasquier et al., 2019; Tan et al., 2023; Ginsburg et al., 2017; Kovac et al., 2016; Spiess et al., 2013; Pisters et al., 2008; Jones et al., 2018; Crouzet et al., 2017; Kanthabalan et al., 2017.

- **What it is for**: every modality clusters at **~50–70% control**, so the choice rests on **morbidity, anatomy and expertise**. **Pattern**: surgery reports **surgical** endpoints, the rest **Phoenix** — not directly comparable; **fistula** is the ablative signature, **incontinence** the surgical one; **impotence** is near-universal. **Trap**: the two pad figures measure **different thresholds** of one outcome — neither is "the" number.

COMPARISON — SALVAGE LOCAL OPTIONS AFTER RADIOTHERAPY

Option	Selection	Advantage	Chief morbidity
Salvage RP	EAU criteria (above)	Longest data; pathological staging	Incontinence, contracture, rectal injury, impotence
Salvage BT	After EBRT only		

Option	Selection	Advantage	Chief morbidity
		Spares irradiated neighbours	GU (stricture, incontinence, haematuria)
Salvage SBRT	After EBRT	Low toxicity	Modest control
Salvage cryotherapy	PSA <10 (ideally <4), doubling time ≥16 months	Focal or whole-gland	Urethrectal fistula ; erectile dysfunction
Salvage HIFU	≥2 years after EBRT	Focal or whole-gland	Fistula, osteitis pubis , stricture, incontinence

Follow-up After Salvage Treatment of Radiation Recurrence

- **Recurrence in more than half** after maximal local therapy. **No guidelines**: periodic **PSA**, with a **modality-appropriate biochemical definition** triggering **restaging imaging**.

After Focal Therapy

- **Treat as a radiation failure**: **advanced imaging** → **MRI** → **targeted biopsy** → **salvage** — **surgery**, **radiation** (EBRT, SBRT, LDR- or HDR-BT) or **ablation** (repeat HIFU or cryotherapy).

KEY POINTS

- Curative options for radiation recurrence: **salvage prostatectomy**, **re-irradiation (BT or SBRT)**, **ablation (cryotherapy or HIFU)**.
- Well selected: **grade group <4** and **pre-salvage PSA <10**.
- **Salvage prostatectomy** is harder than primary surgery (**fibrosis**), with more **incontinence, impotence and complications**.
- **Salvage BT** only after **external RT**; **LDR ≈ HDR**.
- **Ablation** may be **focal or whole-gland**.
- **Focal therapy recurrence** is managed **as radiation recurrence**.

Concept — After radiotherapy every salvage modality buys a similar, moderate chance of cure at the price of morbidity in irradiated tissue; **selection** — *proven local-only disease, low PSA, lower grade, long life expectancy* — predicts success far more than the choice of modality.

NONCURATIVE INTENT APPROACHES TO BIOCHEMICAL RECURRENCE

PAIR — THE GOVERNING TRADE

Observation	Androgen deprivation therapy
Preserves quality of life	Delays metastatic recurrence
Risks metastatic progression and symptoms	Risks side effects and castration resistance (from prolonged suppression)

- **AUA**: **observation or trial**; **ADT only for higher metastatic risk** — **doubling time <12 months** or **high grade**; if used, **intermittent over continuous**.

Observation

Natural History

- Most men **will not die of prostate cancer** even after recurrence. **Pound (untreated post-RP BCR):** only about **a third metastasised** (median **8 years**), dying a median **5 years** later — and fewer than half of those died of prostate cancer. **Pound predictors:** time to BCR **<2 years**, doubling time **<10 months**, Gleason **8–10**. → For many, treating BCR alone **does more harm than good**.
- **AUA: observation over hormones** after maximal local therapy. **EAU:** observation if **unwilling** to have local therapy, **limited life expectancy**, or **EAU low-risk BCR** — **post-RP:** doubling time **>1 year** and grade group **<4**; **post-RT:** interval to failure **>18 months** and grade group **<4**.

Follow-up and Evaluation

- **PSA surveillance and periodic imaging** — AUA prefers **PET/CT**, especially if **doubling time <12 months**.

KEY POINTS

- **Observation** is reasonable because **most men will not die of biochemically recurrent disease**.
- **AUA: observation over hormonal therapy**; **EAU: observation for low metastatic risk**.
- Observed men need **routine surveillance imaging**.

Androgen Deprivation Therapy

Historical Background

- **1893 — White:** castration for urinary obstruction (ageing dogs). **1941 — Huggins and Hodges:** castration improves far-advanced prostate cancer; injected androgens aggravate it → **Nobel Prize 1966**.
- **Enabling discoveries:** testosterone isolated, characterised, synthesised; prostatic acid phosphatase linked to prostatic epithelium and advanced cancer; castration's effect on **benign prostatic hyperplasia (BPH)**.
- **"Studies on Prostate Cancer I–III":** adult prostate = source of acid phosphatase; carcinoma = malignant adult-type epithelium; orchidectomy → **prompt, sustained fall in acid phosphatase**, longer survival, weight gain, rising red cells, pain relief, less leg oedema — at the cost of **lost sexual capacity and hot flushes**.
- **Castration resistance** made it **rarely curative**; Huggins suspected **adrenal androgens** → **bilateral adrenalectomy (1945):** modest, brief responses, severe morbidity.
- **1960s — Veterans Administration Cooperative Urological Research Group (VACURG):** diethylstilboestrol (DES) vs orchidectomy — effective but **cardiovascular and thromboembolic toxicity**.
- **Hypothalamic–pituitary–gonadal (HPG) axis:** oestradiol inhibits luteinising hormone (LH); LHRH isolated (1971) → agonists and antagonists. **Agonist** → **testosterone surge ("flare")** → **loss of pituitary stimulation** → ↓ LH → ↓ **Leydig testosterone**; depot formulations made agonists **dominant**.

The Androgen Receptor

- The **androgen receptor (AR)** is the **key oncogenic driver**; antiandrogens **interrupt ligand binding**. **Trap:** many antiandrogens **block central feedback**, so **alone they raise LH and testosterone**.
- **AR: nuclear receptor superfamily** (sex and adrenal steroids, thyroid, vitamin D, retinoids) — **ligand-inducible transcription factors** with **three domains:** **ligand-binding (LBD)**, **DNA-binding**, **transactivation**. **Steroid diffuses in** → **binds LBD** → **activates DNA binding** → **transactivation domain recruits co-regulators** → **gene repression or activation**. **Gene on the X chromosome**, mapped through cancer and **androgen insensitivity mutants**. **All ADT blocks AR activity** — by **cutting ligand** or **inhibiting the receptor**.

- **Castration resistance** (formerly "androgen-independent"/"hormone-refractory") was thought non-AR-mediated, but **rapid ("warm") autopsies** showed **AR signalling remains necessary and sufficient** → renamed "**castration-resistant**".
- **Mechanisms:**
 1. **AR overexpression/amplification** (~20–30% of castration-resistant prostate cancer [CRPC] metastases) → **hypersensitivity to low androgen**.
 2. **Activating LBD mutations** → **altered specificity**: other steroids and **antiandrogens activate AR** — basis of **antiandrogen withdrawal**.
 3. **Splice variants lacking the LBD** → **ligand-independent AR**; **AR-V7** predicts **castration resistance and poor survival**.
 4. **Bypass pathways**, e.g. **glucocorticoid receptor co-option**.
- **Fig. 165.4: normal** — testosterone → 5 α -reductase → dihydrotestosterone (DHT) → AR-DHT dimer → nuclear translocation → androgen response element (ARE) → PSA, growth, survival; resistance — amplification; mutation and splicing (AR- Δ LBD); intratumoural steroidogenesis; bypass (GR, MAPK, SOX2, WNT, BCL2, AKT).

Molecular Biology and Sources of Androgen

- **Testis: Leydig cells under LH**; testosterone **diffuses freely**. In cells with **AR and 5 α -reductase**, testosterone → DHT, which binds AR ~**13-fold** more avidly (**preferred ligand**) → nucleus → **AREs**.
- **Adrenal: androstenedione and dehydroepiandrosterone (DHEA)**, driven by **adrenocorticotrophic hormone (ACTH)** ← corticotropin-releasing factor — **too weak alone to sustain prostatic epithelium after castration**.
- **Tumour: autocrine steroidogenesis** (raised intratumoural androgens, upregulated enzymes) → rationale for **synthesis inhibitors**.

Concept — Castration removes the ligand, not the receptor. Because the AR remains the driver even in "castration-resistant" disease, every resistance mechanism — amplification, mutation, splicing, local synthesis, bypass — keeps AR signalling alive, and every later drug hits the receptor or its ligand supply harder.

METHODS OF ANDROGEN DEPRIVATION THERAPY

- **Four approaches: orchidectomy; AR antagonism; LHRH/LH inhibition; androgen synthesis inhibition.**
Classic ADT = **chemical or surgical castration**; castrate testosterone <50 ng/dL. Some agents are licensed only in **metastatic or castration-resistant** disease.

TABLE 165.3 HORMONAL INTERVENTIONS AND ENDOCRINE AXIS IN PROSTATE CANCER

Class	Agents	Site	Mechanism	Comments
Gonadotrophin-releasing hormone (GnRH) agonists	Leuprolide, goserelin, triptorelin	Pituitary	Stimulate LH; feedback then lowers LH	Mainstay in metastatic disease; ↑ follicle-stimulating hormone (FSH)
GnRH antagonists	Degarelix, abarelix, cetorelix	Pituitary	Block GnRH receptors	No surge; ↓ FSH
Adrenal-targeting	Ketoconazole, abiraterone, orteronel (TAK-700), galeterone	Adrenal	Block cytochrome P450 steroidogenesis	Selective CYP17 inhibitors with low-dose prednisone; hypertension, oedema, hypokalaemia

Class	Agents	Site	Mechanism	Comments
First/second-generation AR antagonists	Flutamide, nilutamide, bicalutamide	Prostate / cancer	Nonsteroidal ; block testosterone/DHT binding	Monotherapy not indicated; gynaecomastia
Third-generation AR antagonists	Enzalutamide, apalutamide, darolutamide	Prostate / cancer	Block AR, nuclear translocation , transcription	Falls, fatigue, anorexia
5α-reductase inhibitors	Finasteride (type II); dutasteride (I and II)	Prostate / cancer	Block testosterone → DHT	BPH; not for prostate cancer

- **What it is for:** placing each drug at its **level on the axis**. **Pattern:** agonist ↑ FSH, antagonist ↓ FSH. **Traps:** 5 α -reductase inhibitors are not cancer treatment; ketoconazole is non-selective — "selective CYP17" means **abiraterone** and successors.

Orchiectomy

- **~95% of testosterone is testicular:** orchidectomy gives **castrate levels often within 6 hours, <20 ng/dL within 12 hours**, median **~15 ng/dL**, nearly all **<30 ng/dL**; VACURG showed **pain relief and better performance status**.
- **Simple** (all tunical tissue removed) **vs subcapsular** (non-functional tunica kept — avoids an **empty scrotum**; **operator-dependent**, must remove all Leydig tissue; done properly, **equivalent**).
- **Underused and declining** despite **lower cost and equal survival**; versus LHRH agonists, **fewer cardiac events, peripheral arterial disease and fractures**, and with **≥35 months** of therapy **less thromboembolism and diabetes** → offer for lifelong ADT.

Antiandrogens

- **AR antagonists competing at the LBD:** **steroidal** or **nonsteroidal** (first, second, third generation).

Steroidal

- **Cyproterone acetate** (no longer used for cancer): **AR blockade + progestational central suppression** (testosterone ↓ **70–80%**); 100 mg two to three times daily; **inferior to goserelin**; **severe cardiovascular events (up to 10%)**, gynaecomastia, rare **fulminant hepatotoxicity**; still used **50–100 mg/day** for hot flashes. **Medroxyprogesterone** and **megestrol acetate** are the same class — not routine.

Nonsteroidal First and Second Generation

- **Alone they raise LH and testosterone (~1.5× normal)** — antiandrogen action **without hypogonadism**; potency preservation is theoretical, **uncommon in practice**.
- **Class toxicity:** **gynaecomastia/mastodynia** (aromatisation of excess testosterone to oestradiol); **hepatotoxicity** in all (monitor liver tests); **interstitial pneumonitis** → **fibrosis** in all, **highest with nilutamide (1–2%)**.
- **Role:** **combined androgen blockade (CAB)** with LHRH agonists — above all **blocking the flare** for **~30 days**.

COMPARISON — FIRST- AND SECOND-GENERATION NONSTEROIDAL ANTIANDROGENS

Agent	Half-life	Dosing	Signature	Status
Flutamide (1st)	~6 h (2-hydroxyflutamide)	250 mg three times daily	Diarrhoea; first "pure" antiandrogen; OS inferior to DES	Rare
	56 h (hepatic P450)	300 mg daily × 1 month → 150		Historical

Agent	Half-life	Dosing	Signature	Status
Nilutamide (1st)		mg (steady state 14 days)	Delayed dark adaptation (~quarter, reversible); most pneumonitis	
Bicalutamide (2nd)	~6 days	50–150 mg once daily	Most potent, best tolerated; kinetics unaffected by age, renal or moderate hepatic impairment; testosterone stays normal	Only indication: 50 mg + LHRH agent in metastatic disease (CAB or flare)

- **Bicalutamide monotherapy: 50 mg inferior to castration; 150 mg ≈ castration** in locally advanced/metastatic disease with **better sexual interest and physical capacity**, but **gynaecomastia 66%, breast pain 73%**; in **low-risk localised disease worse OS** than watchful waiting (Early Prostate Cancer programme). **Never approved as a single agent.**

Nonsteroidal Third Generation

- **AR-targeting agents (ARTAs)** block ligand binding, nuclear translocation and DNA binding; more potent than earlier agents.

COMPARISON — THIRD-GENERATION ANTIANDROGENS (ARTAS)

Agent	First approval (trial)	Other key trials	FDA indications	Signature toxicity
Enzalutamide 160 mg	2012 — AFFIRM: metastatic CRPC (mCRPC) after docetaxel ; better radiographic PFS and OS	PREVAIL (pre-chemo mCRPC), PROSPER (nmCRPC), ENZAMET , ARCHES (mHSPC), EMBARK (high-risk nmHSPC)	All CRPC (M0/M1, before/after chemo); metastatic hormone-sensitive (mHSPC); high-risk non-metastatic hormone-sensitive (nmHSPC)	Fatigue, breast enlargement, diarrhoea, hot flushes; seizures <1%
Apalutamide 240 mg	2018 — SPARTAN: nmCRPC; MFS 40.5 vs 16.2 months (HR 0.28)	TITAN (mHSPC)	nmCRPC (doubling time ≤10 months); mHSPC	Rash 25%, falls/fractures 12%, hypothyroidism 8%
Darolutamide 600 mg twice daily	2019 — ARAMIS: nmCRPC, doubling time ≤10 months; OS HR 0.68	ARASENS	nmCRPC; mHSPC with docetaxel	Fatigue, falls, fracture, rash, weight gain — lowest rates

- **Traps: seizures → enzalutamide; rash, hypothyroidism → apalutamide; darolutamide in mHSPC only with docetaxel.** Generation labels are loose — **bicalutamide** is called both "second generation" and the most potent "first-generation" agent.

Antiandrogen Withdrawal Syndrome

- On **CAB** with biochemical progression, **stopping the antiandrogen can drop PSA** (sometimes objective responses) — **AR mutations turn the antagonist into an agonist.**
- **Bicalutamide:** response **within 6 weeks**; **15–30%** get **>50% PSA fall**, lasting **3.5–5 months**. With **enzalutamide-class agents 0–10%**, brief, no radiographic response (less agonism). **No OS benefit.**

Inhibition of Luteinizing Hormone–Releasing Hormone

Luteinizing Hormone–Releasing Hormone Agonists

- **Leuprolide, goserelin, triptorelin, histrelin:** chronic receptor exposure → **desensitisation** → **↓ LH** → **↓ testosterone**; potency and half-life from **amino acid substitution at position 6**; **depots (28 days)** to **osmotic implants (1 year)**; **survival equals orchidectomy.**

- **Flare:** LH up to **10-fold** for **10–20 days** → **life-threatening exacerbation** almost only with **spinal/weight-bearing bone metastases**; prevented by **antiandrogen** before or with the agonist (simultaneous start is as good as a week's lead-in), needed only **21–28 days**.

Luteinizing Hormone–Releasing Hormone Antagonists

- **Abarelix, cetrorelix, degarelix, ganirelix, relugolix:** immediate competitive blockade — LH ↓ **84%** in 24 hours, **no flare** (the major advantage) → preferred in **eugonadal men with impending cord compression or severe bone pain** when orchidectomy is unsuitable.
- **Abarelix:** rapid, effective castration; early antagonists caused **histamine reactions**; abarelix can cause **severe allergy even after uneventful doses** → **observe ≥30 minutes**; US use restricted to men **unable to take other hormones who refuse orchidectomy**.
- **Degarelix:** **no systemic allergy**; vs leuprolide castrate (<50 ng/dL) by day 3 in >95% vs none, equal maintenance to 12 months (**non-inferior**); more **injection-site reactions, chills**, fewer **urinary infections, arthralgia**; pooled data favour it, but **no randomised survival comparison**.
- **Relugolix** — oral, FDA 2020. HERO (BCR or newly metastatic, vs leuprolide, 48 weeks): **castrate at day 4 56% vs 0%**; **sustained castration 97% vs 89%**; **testosterone recovery 90 days after stopping 54% vs 2%**. About **half** of men on LHRH agents **never recover baseline testosterone by 5 years**.

PAIR — LHRH AGONIST VERSUS ANTAGONIST

Agonist	Antagonist
Stimulate, then desensitise	Block immediately
Flare → antiandrogen 21–28 days	No flare → cord compression
FSH rises	FSH falls
Slow testosterone recovery	Relugolix: rapid recovery

Inhibition of Androgen Synthesis

COMPARISON — ANDROGEN SYNTHESIS INHIBITORS

Agent	Target	Consequence and replacement	Niche / toxicity
Aminoglutethimide	Cholesterol → pregnenolone	Blocks cortisol and aldosterone as well as DHEA ("medical adrenalectomy") → cortisone + fludrocortisone	CRPC responses with hydrocortisone; rare in PSA era. Anorexia, nausea, rash, lethargy, vertigo, hypothyroidism, nystagmus
Ketoconazole	Non-selective P450; 17,20-desmolase (C21 → C19) in adrenal and Leydig cells	Castrate within 4 hours, immediately reversible → 200–400 mg 8-hourly + hydrocortisone 5–20 mg twice daily	Emergency castration (cord compression) ; abiraterone-naïve second-line (with antiandrogen withdrawal: more responses, no survival gain); testosterone escapes within 5 months without other ADT. Gynaecomastia (how it was discovered), lethargy, weakness, hepatotoxicity , visual disturbance, nausea
Abiraterone	Selective CYP17: 17 α -hydroxylase + C17,20-lyase	↓ cortisol → ↑ ACTH → ↑ deoxycorticosterone/corticosterone → hypokalaemia, hypertension, fluid overload (aldosterone itself held by renin)	mCRPC and mHSPC (Fig. 165.5)

Agent	Target	Consequence and replacement	Niche / toxicity
		feedback) → prednisone 5 mg once or twice daily suppresses ACTH	

- **Abiraterone landmark trials:** COU-AA-301 (mCRPC after docetaxel): PFS **5.6 vs 3.6**, OS **14.8 vs 10.9 months**. COU-AA-302 (pre-chemotherapy mCRPC): radiographic PFS **16.5 vs 8.3**, OS **not reached vs 27.2 months**. LATITUDE (high-risk mHSPC; stopped early): radiographic PFS **33.0 vs 14.8**, OS **not reached vs 34.7 months (HR 0.62)**.
- **What it tests: the more proximal the block, the more replacement** — glucocorticoid and mineralocorticoid (aminoglutethimide), glucocorticoid (ketoconazole), **prednisone to suppress ACTH** rather than replace aldosterone (abiraterone).

Concept — *Each method of ADT attacks one level of the axis — testis, pituitary, adrenal synthesis or receptor — and its side-effect signature follows from what else lives at that level: flare at the pituitary, mineralocorticoid excess at the adrenal, gynaecomastia wherever testosterone is left free to aromatise.*

TIMING OF THERAPY

- In non-metastatic BCR, ADT aims to **prolong life and prevent symptomatic progression**, but not all progress and ADT is **lifelong and toxic**. AUA, EAU and NCCN set **no firm start point**. Questions: **immediate vs delayed; continuous vs intermittent** (metastatic data now largely **historical**).

Immediate Versus Delayed

Results in Clinically Localized Disease

- **Primary ADT** (no local therapy, non-metastatic) is **not recommended**: in elderly men it gave **lower cancer-specific survival and no OS gain** versus conservative management, and elsewhere **no change in mortality** except in **high-risk** disease.

Results in Lymph Node Metastatic Prostate Cancer as Adjuvant Therapy

- **Messing (ECOG) trial** showed immediate ADT benefit after node-positive RP, but: **pre-dates modern imaging; small**; delayed arm treated **only at clinical progression**; a fifth had **detectable PSA** (persistent disease); **bulky nodes** (up to 20). **Low-volume N+ with undetectable PSA: 20–40% cured by surgery alone.**

Results in Locally Advanced or Asymptomatic Metastatic Disease or Disease Not Suitable to Local Treatment

- **MRC trial:** the delayed arm had **more progression-related symptoms**; the OS gain faded with follow-up. **EORTC 30891:** **30% of the delayed arm died without ever needing ADT**; benefit concentrated in **PSA >50 ng/mL or doubling time <12 months**.

Results in Biochemically Recurrent Prostate Cancer

- **Observational data:** ADT timing after RP (adjuvant, PSA 0.4/1.0/2.0, or metastasis) **did not change OS**; deferred ADT gave **similar CSS**; salvage ADT vs observation **no survival difference** — **except doubling time <9 months**.
- **TOAD:** delayed arm waited **≥2 years** unless metastasis, symptoms or **doubling time <6 months**.
- **EMBARK** — **high-risk BCR:** **doubling time ≤9 months and PSA ≥1 ng/mL after RP (± RT) or ≥2 ng/mL above nadir after RT**; **1:1:1 leuprolide / leuprolide + enzalutamide / enzalutamide**; **paused at week 37 if PSA <0.2, restarted at 2 (post-RP) or 5 ng/mL (post-RT)**; **MFS 87.3% / 80.0% / 71.4%**; almost half the

combination arm stayed **off treatment >24 months**; 5-year freedom from PSA progression **97.4%** → **enzalutamide approved for high-risk nmHSPC**.

COMPARISON — IMMEDIATE VERSUS DELAYED ADT TRIALS

Trial	Population	Verdict on immediate ADT
Early Prostate Cancer programme	Localised; bicalutamide 150 mg vs placebo	Harm (worse OS)
EORTC 30846	pN1–3 M0, no local therapy	Delayed not shown non-inferior
Messing (ECOG)	pN+ after RP	OS and PFS benefit (bulky, dated)
MRC	Locally advanced / asymptomatic M1	Cancer-specific survival benefit; OS benefit lost ; fewer symptoms
EORTC 30891	Unfit for local therapy	Small OS benefit ; no cancer-specific or symptom-free benefit
TOAD	Rising PSA	OS benefit (HR 0.55)
EMBARK	High-risk BCR	MFS benefit — combination HR 0.42 , enzalutamide alone HR 0.63 vs leuprolide

- **What it tests:** early ADT helps **only in proportion to risk** — harmful in low-risk localised disease, beneficial with **rapid kinetics or high burden**.

Intermittent Versus Continuous Androgen Deprivation Therapy

COMPARISON — INTERMITTENT VERSUS CONTINUOUS ADT

	Rising PSA after RT (NCIC PR.7)	Newly metastatic (SWOG 9346)
Induction	8 months	7 months
Stop if PSA	<4 ng/mL	≤4 ng/mL (randomised only if reached)
Restart at	10 ng/mL or symptoms	20 ng/mL , baseline, or symptoms
Primary endpoint	OS	Co-primary: OS and 3-month quality of life
OS	8.8 vs 9.1 years — non-inferior	5.1 vs 5.8 years (HR 1.10; 90% CI 0.99–1.23) — inconclusive
Deaths	More cancer deaths intermittent; more non-cancer deaths continuous	—
Quality of life	Better hot flushes, sexual desire, urinary symptoms	Better erection and mental health at 3 months only
Verdict	Intermittent acceptable	Intermittent may be worse

- **Why SWOG 9346 is inconclusive:** upper CI **1.23 > margin 1.20** → non-inferiority unproven; lower CI **0.99** includes 1 → inferiority unproven.
- **In practice:** **attrition rises** each cycle (few stop by the first interval, most by the third) and **off-treatment intervals shorten** (~20 → 13 → 9 → 4–5 months). **CRPC** (on ADT with clinical progression or **3 monthly PSA rises**) was treated with **continuous ADT**. **No consensus schedule**. Modern M1 care uses **lifelong combination ADT**, leaving intermittent ADT for men **unsuited to combinations**.

Supportive Care on Androgen Deprivation Therapy

- Discuss side effects at the outset; **mitigate proactively**.

Osteoporosis

- **Androgen suppression (not AR antagonism) lowers bone mineral density (BMD) via loss of peripheral oestrogen. Over half of older men are osteopenic/osteoporotic before ADT (osteoporosis: T-score >2.5 SD below the young-adult mean). Fracture risk ↑ 20–45%, rising with duration.**
- **Baseline hip dual-energy x-ray absorptiometry (DEXA) for anticipated long-term ADT; FRAX (WHO) for risk; testing is badly under-used.**
- **All: calcium 1000–1200 mg/day, vitamin D₃ 400–1000 IU/day, weight-bearing exercise, stop smoking.**
- **Drug treatment if T-score –1.0 to –2.5 with 10-year hip fracture >3% or major fracture >20%: denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly, or alendronate 70 mg orally weekly; repeat DEXA at 1 year.**

Hot Flashes (Vasomotor Symptoms)

- **Commonest side effect (50–80%): upper-body warmth then sweating, from disturbed hypothalamic oestrogen signalling → adrenergic thermoregulatory activation; triggered by posture, hot drinks, ambient temperature; wane but may persist. Treat only if bothersome.**

COMPARISON — TREATMENTS FOR HOT FLUSHES

Class	Agent and dose	Evidence / caveat
Oestrogens	Low-dose DES; transdermal oestradiol	Most effective (up to 90%); gynaecomastia, thromboembolism
Progestogens	Megestrol 20 mg twice daily (→ 5 mg twice daily); cyproterone 50 → 300 mg/day; medroxyprogesterone 20 mg/day	Effective; megestrol stimulates appetite
Non-hormonal	Venlafaxine (serotonin–noradrenaline reuptake inhibitor) 12.5 mg twice daily, 37.5 → max 225 mg/day	Helps over half; less effective than hormonal agents head-to-head; taper — never stop abruptly
	Gabapentin up to 900 mg	Moderate
	Clonidine (central α-agonist), transdermal	Ineffective
Behavioural	Acupuncture / soy / cognitive–behavioural therapy	Uncontrolled / failed / short-lived

- **Pattern:** efficacy and toxicity both run **oestrogen > progestogen > non-hormonal** → progestogens only when flushes significantly impair quality of life.

Sexual Dysfunction (Erectile Dysfunction and Loss of Libido)

- **Profound (Huggins: desire and erection "absent") but not inevitable: ~1 in 10 to 1 in 6 keep erections; libido suffers more (~1 in 20 keep high interest). Penile shortening, volume loss, loss of nocturnal tumescence.**
- **Loss of libido is very hard to treat; phosphodiesterase-5 inhibitors usually fail; intraurethral therapy, vacuum device, intracavernosal injection, penile prosthesis work in selected men; exercise raised sexual interest; sex therapy for couples.**

Cognitive Function

- Hypogonadism worsens **memory and visuospatial** performance (verbal knowledge, mental status, mood spared). Evidence **conflicts**: a small randomised study showed decline in half at 6 months; a **larger controlled study showed none** to 3 years; functional MRI shows **reduced right parieto-occipital activation**. ADT–dementia link is methodologically weak. **Monitor and refer** if decline.

Changes in Body Habitus

- **↑ Appetite**, **↑ insulin** and **abdominal girth**; after a year **weight ↑ ~2–4%**, mostly **fat (↑ ~9–24%)**, with progressive **sarcopenia**; **endurance and grip strength fall within 3 months**.

Diabetes and Metabolic Syndrome

- **Metabolic syndrome in more than half** on long-term ADT — but **atypical: subcutaneous (not visceral) fat and raised HDL cholesterol**, unlike classic visceral, low-HDL syndrome. **↓ Insulin sensitivity**; diabetes risk **↑ with GnRH agonists (HR 1.44) and orchidectomy (HR 1.34)**; diabetics need **more drugs**, worse **HbA1c**. A recent meta-analysis found the metabolic-syndrome link **lost after adjustment**.

Cardiovascular Morbidity and Mortality

- **Raised risk, especially in lower-risk disease**: GnRH agonists **↑ coronary disease, myocardial infarction, sudden cardiac death**, plus **peripheral arterial disease and venous thromboembolism**; **≥6 months** lowers **cardiorespiratory fitness**. **Exception**: no cardiovascular mortality excess with **RT + ADT for locally advanced disease**.

Gynecomastia

- **Gynaecomastia/mastodynia**, together or apart: **DES ~half vs leuprolide rare**; **orchidectomy/agonists ≤5%**; **bicalutamide 150 mg or enzalutamide ~half** (aromatisation).
- **Prevent**: prophylactic breast RT **10–15 Gy single fraction** (useless once established); **tamoxifen 10 mg/day** (non-standard). **Established**: **liposuction or endoscopic subcutaneous mastectomy**.

Anemia

- **Normochromic normocytic, very common** (most on CAB lose **≥10% haemoglobin**; risk **~tripled**, within **6 months**, chronic), even **without marrow disease**; **pre-ADT haemoglobin <12 g/dL predicts shorter survival**; symptomatic in few; **reversible within about a year** of stopping.

Concept — *The case for early ADT strengthens only as the risk of cancer death rises — rapid doubling, high PSA, high burden — because every month of ADT carries a fixed metabolic, skeletal, cardiovascular and sexual cost that the low-risk man pays without a matching gain.*

FOLLOW-UP ON ANDROGEN DEPRIVATION THERAPY (NONMETASTATIC)

- **Aims**: monitor response/progression and **supportive care**; review **3–6-monthly** with **blood count, chemistry, PSA**.
- **Magnitude and speed of PSA response predict durability**: rapid **pre-ADT rise** predicts cancer death; **faster time to undetectable (<0.2 ng/mL)** predicts better survival; undetectable nadir is prognostic in metastatic disease too.
- **HSD3B1 1245A>C** (germline, 15–35%): encodes **3β-hydroxysteroid dehydrogenase-1**, converting **adrenal androgens toward DHT**; the variant **resists degradation** → **excess intratumoural DHT**; **PFS, MFS and OS worsen with each variant allele**; **homozygous ~doubles metastasis rate** — a **predictive biomarker of ADT resistance**, no clinical test yet.
- **Rising PSA** → check **testosterone (<50 ng/dL)**; lower levels within the castrate range may help. **Inadequate suppression** → switch **LHRH agent**, add **oestrogen, antiandrogen or synthesis inhibitor**, or **orchidectomy**.
- **Rising PSA with suppressed testosterone = castration resistance** → image: **axial soft tissue (PET/CT, CT or MRI) + bone (bone scan or PET/CT)**.

KEY POINTS

- ADT: **orchidectomy, AR antagonism, LH suppression, or synthesis blockade.**
- **Rapidity and magnitude of PSA fall predict duration and efficacy.**
- PSA change on ADT → **repeat PSA and testosterone.**
- **Rising PSA with castrate testosterone = CRPC → metastatic evaluation.**
- **ADT's quality-of-life impacts must be managed.**

Concept — On ADT the testosterone level is the referee: a rising PSA means nothing until testosterone is known — inadequate suppression is a drug problem; adequate suppression means the disease has become castration-resistant.

HORMONE SENSITIVE METASTATIC PROSTATE CANCER

- **Metachronous** (after local treatment) vs **synchronous** (at diagnosis). **Metastasis arising on hormonal suppression = CRPC** (next chapter, with nmCRPC). **mHSPC** = metastatic, **no prior hormones.**
- **CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.** Care is **multidisciplinary.**

Diagnosis

- **Conventional (CT, MRI, bone scan) or PET/CT.** Treat what is found — but trials used **conventional imaging**, so **PET-only metastatic disease** is largely unstudied.
- **Histology** not always needed (especially metachronous); **biopsy prostate or metastasis** if never proven.
- **Germline and somatic testing for all metastatic men (AUA, NCCN)** — **>10% carry DNA-repair mutations** with therapeutic implications.

GENETIC TESTING PANEL IN METASTATIC PROSTATE CANCER

Germline	Somatic
BRCA1, BRCA2, ATM, PALB2, CHEK2	BRCA1, BRCA2, ATM, PALB2, CHEK2
HOXB13	FANCA, RAD51D, CDK12
MLH1, MSH2, MSH6, PMS2	Microsatellite instability-high (MSI-H), mismatch repair deficiency (dMMR)

- **Pattern: homologous-recombination core in both; HOXB13 and individual mismatch-repair genes germline-only; FANCA, RAD51D, CDK12, MSI-H/dMMR somatic-only.**

Prognosis

- Median OS **~30 months historically → >48 months** with docetaxel, advanced hormones and combinations.
- **Metachronous low-volume (~92 months) vs synchronous or high-volume (~43 months).**
- **Site (pre-combination era): nodes only > bone > visceral > bone + visceral.**
- **SWOG 9346 PSA nadir at 7 months: ≤0.2 → 75 months; 0.2–4 → 44; >4 → 13.**

Management of Metastatic Hormone Sensitive Prostate Cancer

Systemic Therapy

- **Backbone:** orchiectomy or LHRH agonist/antagonist. Emergency (impending cord compression) → antagonist or orchiectomy. Bone metastases + agonist → add nonsteroidal antiandrogen for the flare. Antiandrogen alone — not recommended; long-term CAB — AUA against (no meaningful survival gain; better combinations exist).
- Unless contraindicated, add docetaxel, an androgen synthesis inhibitor, or an ARTA; growing evidence for "triplet therapy" (ADT + docetaxel + abiraterone or ARTA).
- **Observation not recommended** except asymptomatic men with life expectancy <5 years, or while receiving metastasis-directed therapy for oligometastases.

Docetaxel

- Microtubule inhibitor, 75 mg/m² every 21 days (6 cycles; up to 9 in GETUG-AFU 15). Meta-analysis of the three trials: HR 0.77 — 94 fewer deaths per 1000. Toxicity: grade 3–4 neutropenia, febrile neutropenia, liver dysfunction. Benefit greatest in high volume.

Androgen Synthesis Inhibitor

- Abiraterone 1000 mg + prednisone 5 mg — LATITUDE (≥2 of: grade group 4–5 [Gleason ≥8], ≥3 bone lesions, visceral) and STAMPEDE arm G (M1 plus selected high-risk locally advanced).

Androgen Receptor–Targeting Agents

- Enzalutamide and apalutamide approved with ADT (ARCHES, ENZAMET, TITAN); benefit regardless of volume or prior docetaxel.

"Triplet Therapy"

- PEACE-1 (synchronous; standard care evolved from ADT to ADT + docetaxel; ± abiraterone ± prostate RT, 2 × 2) and ARASENS (ADT + docetaxel ± darolutamide 600 mg twice daily): both improved OS with similar toxicity across arms.

TABLE 165.4 LANDMARK PHASE III CLINICAL TRIALS OF COMBINATION THERAPY IN METASTATIC HORMONE-SENSITIVE PROSTATE CANCER

Trial	Population	Intervention vs control	Primary result (table)	Body text, where different — and why	Key secondary
GETUG-AFU 15	mHSPC	ADT + docetaxel vs ADT	OS not superior: HR 0.88 (0.68–1.14)	HR 1.01 (median OS 58.9 vs 54.2 months) — earlier data cut	Better biochemical and radiographic PFS
CHAARTED (ECOG 3805)	Mainly high volume	ADT + docetaxel vs ADT	OS HR 0.61 (57.6 vs 44.0 months)	—	Benefit only in high volume (49.2 vs 32.2 months)
STAMPEDE (docetaxel)	61% M1, 15% N+, 24% NOMO	ADT + docetaxel + prednisolone vs ADT	OS HR 0.78	—	M1 OS HR 0.76; failure-free survival HR 0.61
STAMPEDE arm G	52% M1, 20% N+, 28% NOMO	ADT + abiraterone + prednisolone vs ADT	OS HR 0.63 (whole trial)	HR 0.61 — M1 subgroup only	3-year OS 83% vs 76%; failure-free survival HR 0.29
LATITUDE	High-risk mHSPC			—	

Trial	Population	Intervention vs control	Primary result (table)	Body text, where different — and why	Key secondary
		ADT + abiraterone + prednisone vs ADT + placebo	OS HR 0.62; radiographic PFS HR 0.47		Pain, chemotherapy, PSA PFS, skeletal events
ARCHES	~62% high volume	ADT + enzalutamide vs ADT + placebo	Radiographic PFS HR 0.39	—	PSA progression, undetectable PSA, response
ENZAMET	52% high volume; ~45% early docetaxel	ADT + enzalutamide vs ADT + standard nonsteroidal antiandrogen	OS HR 0.67 (3-year 80% vs 72%)	—	Biochemical and clinical PFS
TITAN	61% high volume; ~10% prior docetaxel	ADT + apalutamide vs ADT + placebo	Radiographic PFS HR 0.48; OS HR 0.67	—	Time to chemotherapy
PEACE-1	Synchronous, ~64% high volume	ADT ± docetaxel ± RT + abiraterone vs same without	Radiographic PFS HR 0.54; OS HR 0.82 (all)	HR 0.50 and 0.75 — docetaxel-treated subgroup	Abiraterone adds to ADT + docetaxel
ARASENS	mHSPC	ADT + docetaxel + darolutamide vs ADT + docetaxel + placebo	OS HR 0.68	—	Castration resistance, pain, skeletal events, next therapy

- **What it is for:** ADT alone is no longer standard — every intensification class improves outcomes. **Pattern:** docetaxel depends on volume; abiraterone, enzalutamide, apalutamide do not; triplets add a hormonal agent on top of docetaxel. **Traps:** GETUG-AFU 15 is the negative docetaxel trial; ENZAMET's control had an older antiandrogen, not placebo.

COMPARISON — LATITUDE HIGH RISK VERSUS CHAARTED HIGH VOLUME

	LATITUDE high risk	CHAARTED high volume
Criteria	≥2 of: Gleason ≥8, ≥3 bone lesions, visceral	≥4 bone metastases (≥1 beyond axial skeleton) or any visceral
Includes grade?	Yes	No
Selects for	Abiraterone (entry)	Docetaxel (benefit subgroup)

Local Therapy to the Primary

- In **synchronous** disease the untreated primary may seed metastasis.

COMPARISON — RADIOTHERAPY TO THE PRIMARY IN MHSPC

Trial	Overall OS	Low volume	Verdict
HORRAD (ADT ± prostate RT)	No difference (HR 0.9)	Suggested benefit <5 sites (HR 0.68, not significant)	Hypothesis-generating
STAMPEDE arm H (pre-specified low-volume analysis)	No difference (HR 0.92)	Improved: HR 0.68; 3-year OS 81% vs 73%	RT to primary in low-volume synchronous disease

- **Caveat:** these men had **ADT only**; benefit on top of modern combinations is unknown.

- **Cytoreductive prostatectomy: feasible** (low morbidity, good continence) — **trials or palliation of local symptoms only.**

Metastasis-Directed Therapy

- PET/CT detects **oligometastatic disease (≤ 3 sites)** targetable by **surgery or stereotactic RT** — a **non-hormonal** strategy.
- **Salvage lymph node dissection:** about half free of clinical recurrence at 2–3 years, but **long-term BCR-free survival is low** while **OS stays high.**

COMPARISON — RANDOMISED PHASE II TRIALS OF METASTASIS-DIRECTED THERAPY (MDT)

Trial	Imaging	Arms	Primary endpoint	Result
STOMP	^{11}C -choline, ≤ 3 lesions	Surgery or SBRT vs observation	ADT-free survival	21 vs 13 months
ORIOLE	PSMA	SBRT vs observation	Progression at 6 months	19% vs 61%
Pooled	—	—	PFS	11.9 vs 5.9 months (HR 0.44)

- **What it tests:** MDT **delays** systemic therapy; **no OS data** → **highly motivated men wishing to avoid lifelong therapy.**

KEY POINTS

- ADT + docetaxel improves survival, driven by **high-volume** disease.
- ADT + abiraterone, enzalutamide or apalutamide improve survival **regardless of volume or prior docetaxel.**
- ADT + docetaxel + abiraterone or darolutamide improves survival; **triplets** are for the **chemotherapy-fit**, by shared decision on **volume** (high benefits more), **synchronous vs metachronous** (synchronous benefits more) and preference.
- **Local therapy to the primary** may help **low-volume synchronous** disease.
- MDT may prolong **systemic-therapy-free survival** in oligometastatic disease.

Concept — *In mHSPC, ADT is the floor, never the ceiling: every randomised addition has improved outcomes, and what remains is judging burden — chemotherapy where volume and synchronous presentation justify it, radiotherapy to the primary only where volume is low, metastasis-directed therapy only where the lesions are few enough to count.*

FOLLOW-UP WHILE ON THERAPY FOR METASTATIC HORMONE SENSITIVE PROSTATE CANCER

- AUA, EAU, NCCN: review **3–6-monthly** — **PSA and treatment-relevant bloods (liver, blood count).** Testosterone with any PSA rise (rising PSA + castrate testosterone = CRPC). Image only if management will change or CRPC is suspected. Maintain **bone health (baseline and periodic DEXA)** and **cardiovascular health.**